

Assignment 3: Ring Oscillator Design and Analysis

5-Stage CMOS Ring Oscillator in SCL 180 nm Technology

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Note. This report was assembled from Cadence screenshots (schematic, CDF parameter dialogs, transient plots, and the parametric-sweep marker readouts). A few places required either an assumption or data that was not present in the screenshots provided; every such place is explicitly flagged as a note at the point it occurs, and all of them are collected again in Section 7 for a final check before submission.

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1 Introduction and Design Specifications

A ring oscillator is formed by connecting an odd number of inverting gain stages in a closed loop, with the output of the last stage fed back to the input of the first. Because the loop contains an odd number of inversions, it has no stable DC operating point; small perturbations (device noise, simulator initial conditions) grow and propagate around the loop, and the finite propagation delay of each stage sets the frequency at which the loop self-sustains oscillation.

Ring oscillators are one of the most widely used test structures in digital and mixed-signal IC design. Beyond being a compact way to characterize the raw speed of a fabrication process (a "speed monitor" placed on almost every test chip), they are used as:

- the free-running core oscillator inside an on-chip PLL or DLL, before it is locked to a reference clock;
- a fast, low-area entropy source for on-chip random number generation, since the jitter (cycle-to-cycle period variation) of an unlocked ring oscillator is dominated by thermal and flicker device noise;
- a voltage- or temperature-sensitive delay reference, exploited deliberately in this assignment's Part 3, where the strong dependence of f_{osc} on V_{DD} is used to characterize how delay scales with supply.

This assignment designs, simulates, and characterizes a 5-stage ring oscillator built from the same CMOS inverter (same transistor sizing) designed in the previous assignment, in the SCL 180 nm process.

Table 1: Design specifications

Parameter	Value
Technology	CMOS SCL 180 nm (<code>ts018_scl_prim</code>)
Supply voltage, V_{DD} (nominal)	1.8 V
Number of stages, N	5 (odd)
Channel length (NMOS & PMOS), L	0.18 μm (minimum)
NMOS sizing (per stage)	$W_n = 0.5 \mu\text{m}$, $\text{nf} = 1$
PMOS sizing (per stage)	$W_p = 1.5 \mu\text{m}$, $\text{nf} = 5$ (finger width 0.3 μm)
Load capacitance per stage, C_L	30 fF
Simulator	Cadence Spectre (Virtuoso ADE-L)
Analysis type	<code>tran</code> , 0 to 10 ns, <code>moderate</code> accuracy

The transistor sizing above is carried over unchanged from the CMOS inverter designed in the previous assignment, as instructed. This gives

$$\frac{(W/L)_p}{(W/L)_n} = \frac{1.5/0.18}{0.5/0.18} = 3.0 \quad (1)$$

i.e. the PMOS is sized roughly $3\times$ wider than the NMOS to compensate for the lower hole mobility relative to electron mobility in this process. This is the same ratioed-sizing idea used to bring the switching threshold V_M of a single inverter close to $V_{DD}/2$, and it is reused here without re-optimization, since the assignment explicitly asks for the same sizing to be carried forward.

2 Q1: Circuit Design of the 5-Stage Ring Oscillator

2.1 Theory

2.1.1 The Barkhausen criterion for a ring of inverters

A feedback loop sustains oscillation at a frequency f_0 if the Barkhausen criteria are simultaneously satisfied at f_0 :

1. **Phase condition:** the total loop phase shift is 0° (or an integer multiple of 360°).
2. **Gain condition:** the loop gain magnitude is at least unity, $|A_{loop}(f_0)| \geq 1$.

Each CMOS inverter stage contributes a DC phase shift of 180° (it is inverting) plus an additional, frequency-dependent phase lag caused by its finite propagation delay t_{pd} : charging or discharging the capacitance at its output node is not instantaneous, so the output transition always lags the input transition by a small time delay, which corresponds to a phase lag that grows with frequency. For N identical stages in a loop, the total DC phase is $N \times 180^\circ$. When N is *odd*, this is an odd multiple of 180° – never 0° or 360° – so the loop has no stable DC operating point and is forced to keep switching indefinitely. The loop settles into steady-state oscillation once the additional delay-based phase lag accumulated by all N stages reaches the remaining 180° needed to complete a full 360° round trip – equivalently, once the signal has propagated all the way around the loop *twice* (once to invert the logic level, once again to return it to the original polarity). This gives the well-known ring-oscillator period and frequency relations:

$$T = 2Nt_{pd}, \quad f_{osc} = \frac{1}{2Nt_{pd}} \quad (2)$$

where t_{pd} is the average propagation delay of a single stage (averaged over its high-to-low and low-to-high delays, since the two are generally not equal).

An *even* number of stages, by contrast, gives $N \times 180^\circ = 0^\circ \pmod{360^\circ}$ at DC, which *is* a stable operating point – the circuit becomes a bistable latch (it settles into one of two fixed logic states) rather than an oscillator. This is why N must be odd, and why this design uses $N = 5$.

2.1.2 Gain condition and large-signal behaviour

The gain condition is comfortably satisfied here: a CMOS inverter loaded only by the next stage's gate capacitance and an explicit C_L has a very large small-signal gain in its transition region (tens of dB, set by the transconductances and output resistances of the NMOS/PMOS in saturation near V_M), far in excess of the unity gain the Barkhausen criterion requires. This means the "sinusoidal small-signal oscillator" picture (where the gain condition is the binding constraint, e.g. in an LC or crystal oscillator) is not really the operating regime here. Instead, each stage is driven hard into either the fully-on or fully-off large-signal region on every cycle, and the oscillation amplitude simply saturates at the supply rails ($\approx 0\text{ V}$ and $\approx V_{DD}$) rather than settling at some intermediate small-signal value. The resulting waveform is consequently a near rail-to-rail square wave (visible in Fig. 6) rather than a sinusoid, and the oscillation frequency is determined almost entirely by the large-signal switching delay t_{pd} , not by a small-signal pole/zero analysis.

2.1.3 Startup transient

Because the loop is only marginally unstable near the (unbiased) DC operating point and grows from whatever small perturbation the simulator's initial condition provides, the very first cycle typically does not reach full rail-to-rail swing immediately – the oscillation amplitude builds up over the first one or two cycles before settling into a steady periodic waveform. This "ramp-up" is visible at the start of the simulated transient response (Fig. 6), where the first output pulse does not go all the way down to 0 V on its first excursion; by roughly the second or third period, the waveform has settled into its steady-state, repeatable square-wave shape, and all frequency/power measurements in this report were taken over the full simulated window (which is dominated by this steady-state region).

2.2 Transistor Sizing

Table 2 repeats the per-transistor sizing used for every one of the 5 inverter stages, read directly off the CDF parameter dialogs in the Cadence schematic.

Table 2: Transistor sizing used in every stage of the ring oscillator (identical across all 5 stages)

Device	Total W (μm)	L (μm)	Fingers (nf)	Finger width (μm)	W/L
NMOS (M0, M3, M6, M9, ...)	0.5	0.18	1	0.5	2.78
PMOS (M1, M4, M7, M10, ...)	1.5	0.18	5	0.3	8.33

Each of the 5 output nodes is loaded with an explicit capacitor $C_0 \dots C_4 = 30 \text{ fF}$, representing the fan-out capacitance of the next stage plus any routing parasitics that the assignment lumps into a single explicit load. The supply is a DC source $V_6 = 1.8 \text{ V}$. All five stages use identical device sizing, so the ring is symmetric and every stage is expected to contribute (approximately) equal delay, making the simple $T = 2Nt_{pd}$ relation in Eq. (2) directly applicable without needing to track per-stage mismatches.

Figures 1–3 show the Cadence CDF (Component Description Format) property dialogs for one representative PMOS, NMOS, and load capacitor instance, confirming the sizing values quoted in Table 2 directly at the source.

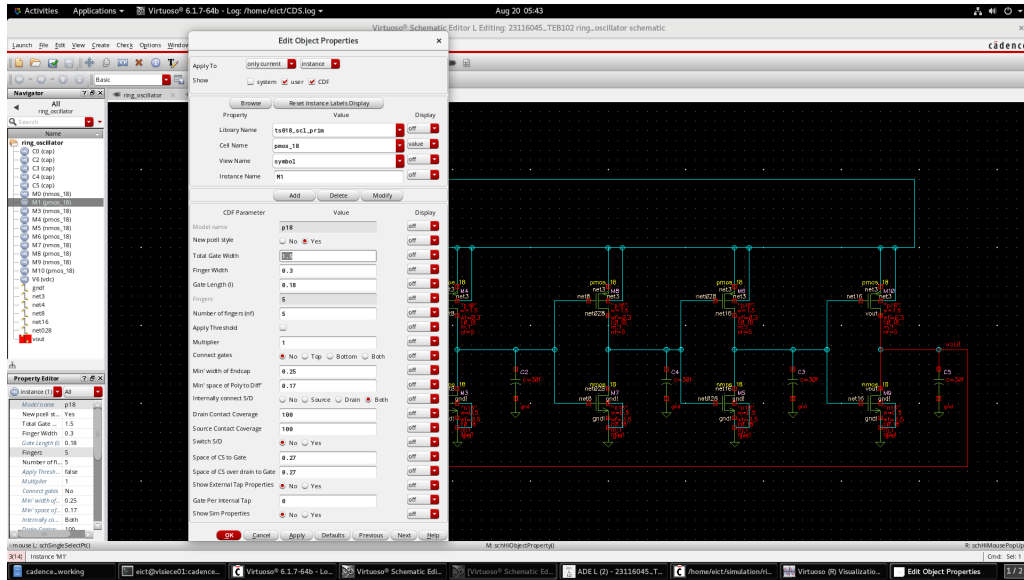


Figure 1: CDF parameter dialog for a representative PMOS instance (M1): Total Gate Width = $1.5 \mu\text{m}$, Gate Length = $0.18 \mu\text{m}$, Number of fingers = 5 (finger width $0.3 \mu\text{m}$).

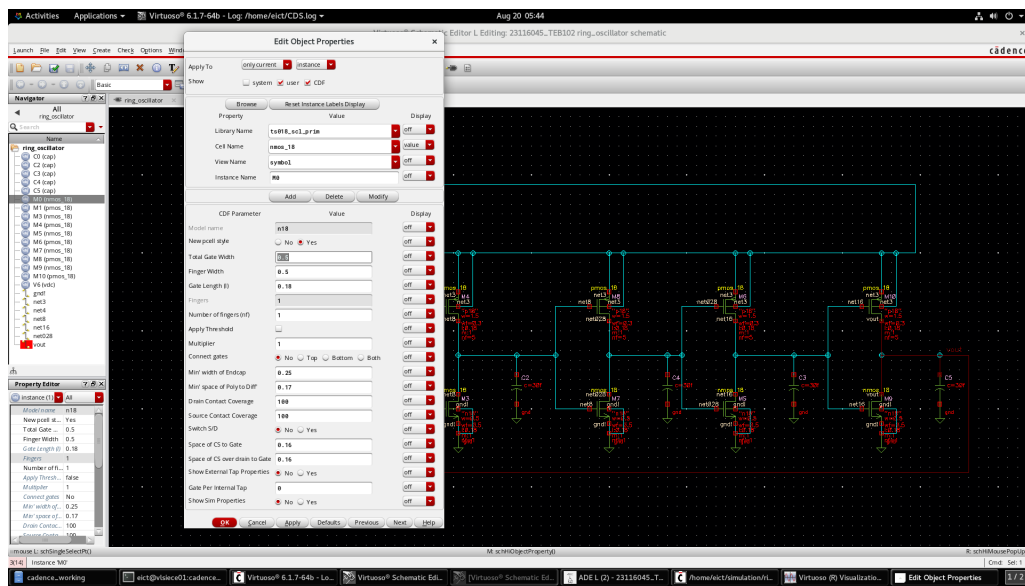


Figure 2: CDF parameter dialog for a representative NMOS instance (M0): Total Gate Width = $0.5\ \mu\text{m}$, Gate Length = $0.18\ \mu\text{m}$, Number of fingers = 1.

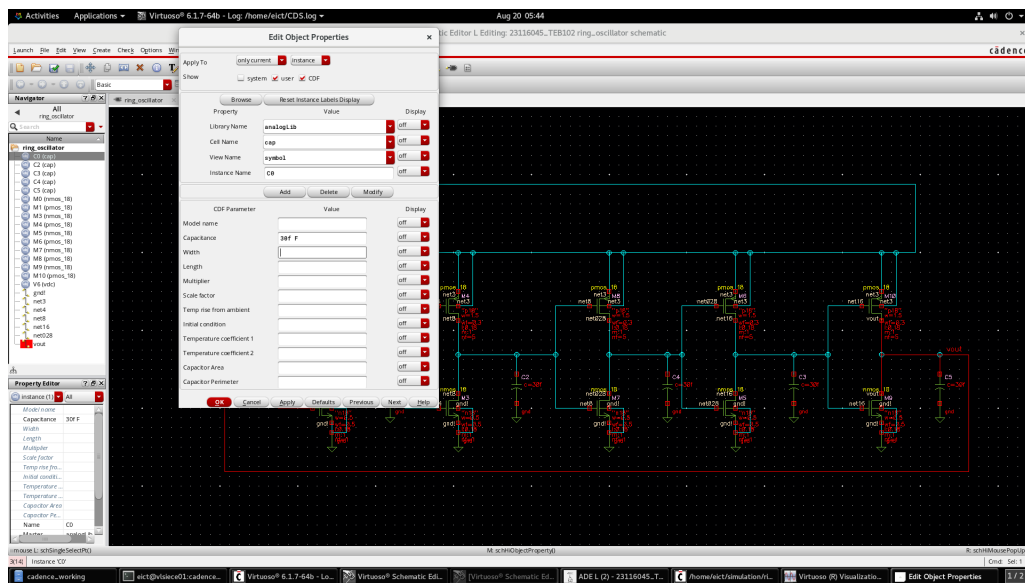


Figure 3: CDF parameter dialog for a representative load capacitor instance (C0): Capacitance = $30\ \text{fF}$.

2.3 Schematic

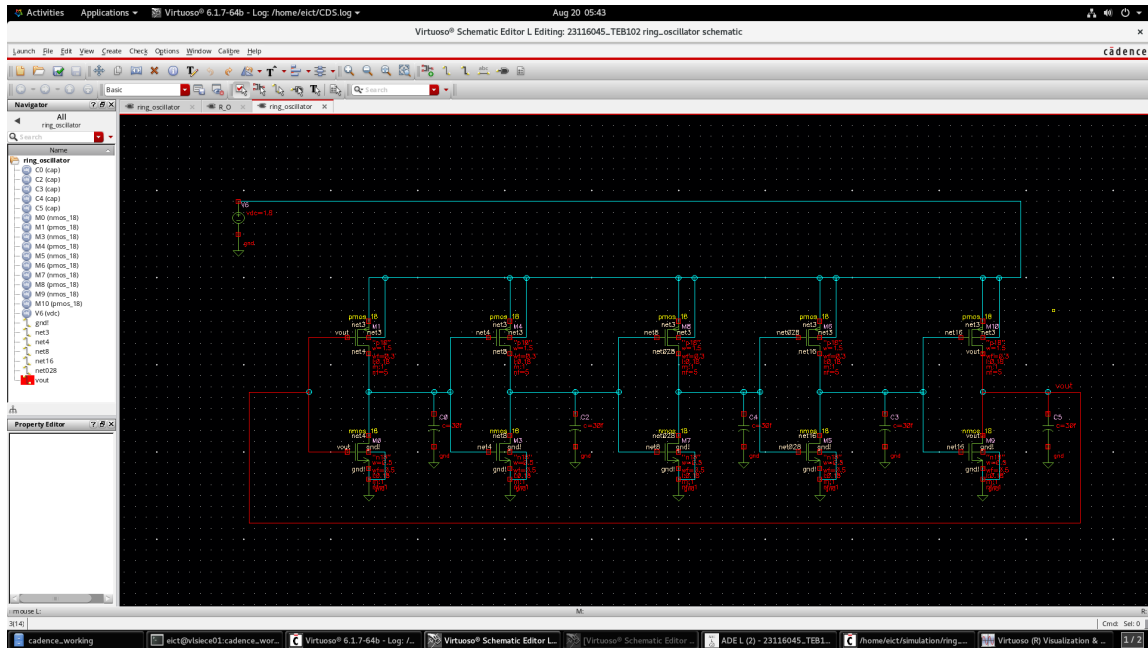


Figure 4: 5-stage ring oscillator schematic in Cadence Virtuoso: 5 inverter stages (M0/M1, M3/M4, M6/M7, M9/M10, and the fifth pair) connected in a closed loop, each output node loaded with a 30 fF capacitor, supplied from $V_{DD} = 1.8\text{V}$.

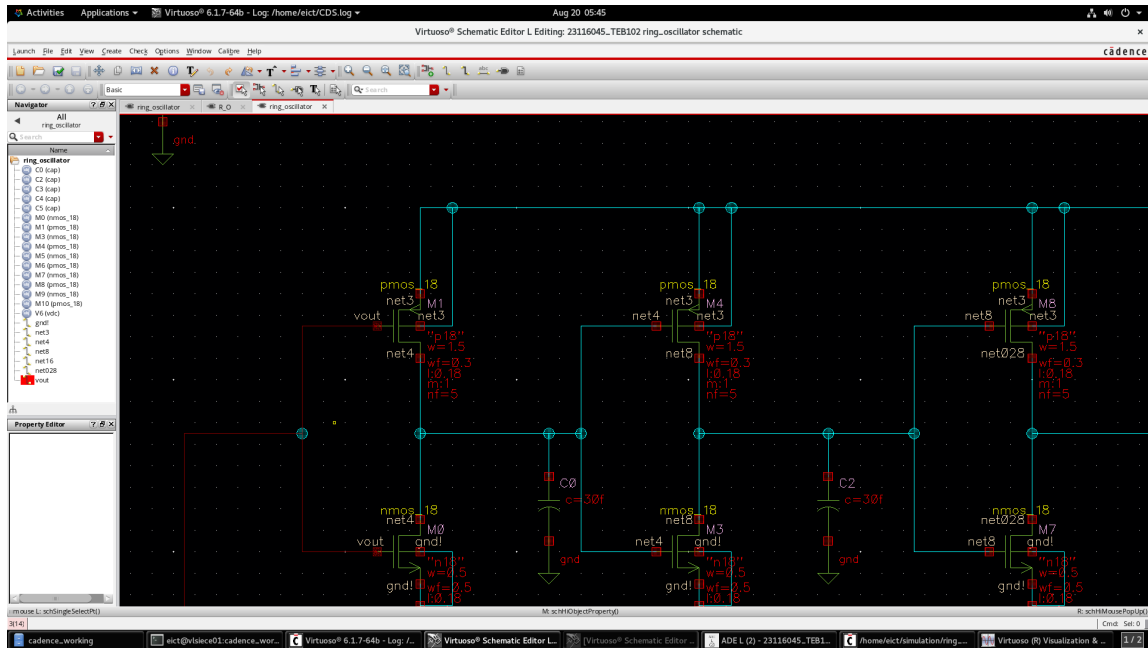


Figure 5: Zoomed-in view of the first three stages of the ring, showing the individual PMOS/NMOS device labels (net3, net4, etc.), the per-instance w , wf , l , nf annotations, and the load capacitors C_0 , C_2 at their respective output nodes.

3 Q2 (Part 1): Theoretical Frequency Calculation

3.1 Delay model

The stage delay t_{pd} in Eq. (2) is estimated with the standard first-order switch-resistance (RC) model taught for CMOS delay estimation. Each transistor is modeled, while it is ON, as a resistor whose value is obtained by linearizing the drain current around $V_{DS} \approx 0$ at a fixed gate overdrive $V_{GS} = V_{DD}$:

$$R_{on} \approx \frac{1}{k' (W/L) (V_{DD} - V_{TH})}, \quad k' = \mu C_{ox} \quad (3)$$

The output node then behaves, to first order, like a simple $R_{on}C_L$ low-pass step response. For a step input, the output of an RC network reaches 50% of its final value at $t = \ln(2) RC \approx 0.69 RC$, and since propagation delay is by convention measured at the 50% crossing, this gives

$$t_{pd} \approx 0.69 R_{on} C_L \quad (4)$$

Because the NMOS pulls the output low through $R_{on,n}$ (giving t_{pHL}) and the PMOS pulls it high through $R_{on,p}$ (giving t_{pLH}), the average single-stage delay used in Eq. (2) is

$$t_{pd} = \frac{t_{pHL} + t_{pLH}}{2} = \frac{0.69 R_{on,n} C_L + 0.69 R_{on,p} C_L}{2} \quad (5)$$

As a cross-check, an alternative constant-current (rather than RC) delay estimate is also computed. This model assumes the transistor stays saturated and delivers a constant current $I_{D,sat}$ throughout the transition (rather than the current tapering off as V_{DS} falls, as the RC model implicitly assumes once the device enters the triode region), so that C_L is charged or discharged approximately linearly:

$$I_{D,sat} = \frac{1}{2} k' (W/L) (V_{DD} - V_{TH})^2, \quad t_{pd} \approx \frac{C_L V_{DD}}{2 I_{D,sat}} \quad (6)$$

(the factor of 2 in the denominator reflects that the 50%-point delay, not the full swing, is being estimated). These two models bracket the delay estimate between two different simplifying assumptions about how current behaves during the transition, and are used here purely as independent sanity checks on each other.

Note – assumed technology parameters. The available screenshots do not include the SCL 180 nm BSIM model-parameter listing (threshold voltage, μC_{ox} , velocity-saturation index, etc.), so an authoritative numeric theoretical value cannot be pulled

directly from the simulation data. The calculation below therefore uses commonly cited *typical* 180 nm textbook figures, purely to demonstrate the method:

$$k'_n = \mu_n C_{ox} \approx 270 \text{ } \mu\text{A}/\text{V}^2, \quad k'_p = \mu_p C_{ox} \approx 90 \text{ } \mu\text{A}/\text{V}^2, \quad V_{THn} \approx |V_{THp}| \approx 0.45 \text{ V}$$

If exact numeric agreement is graded, replace these with the exact values from the PDK's model file (in Cadence: *Tools* $\rightarrow$ *Model File Manager*, or inspect the `.scs/.lib` model card for the n18/p18 devices) and re-evaluate Eqs. (3)–(6).

3.2 Numerical estimate

Using $(W/L)_n = 2.78$, $(W/L)_p = 8.33$, $V_{DD} = 1.8 \text{ V}$, $C_L = 30 \text{ fF}$, and the assumed parameters above:

RC model. From Eq. (3):

$$R_{on,n} = \frac{1}{(270 \times 10^{-6})(2.78)(1.8 - 0.45)} = \frac{1}{(270 \times 10^{-6})(2.78)(1.35)} \approx 988 \text{ } \Omega$$

$$R_{on,p} = \frac{1}{(90 \times 10^{-6})(8.33)(1.8 - 0.45)} = \frac{1}{(90 \times 10^{-6})(8.33)(1.35)} \approx 988 \text{ } \Omega$$

The near-exact equality of $R_{on,n}$ and $R_{on,p}$ is not a coincidence: it is the direct consequence of sizing the PMOS $\approx 3\times$ wider than the NMOS to compensate for $k'_n/k'_p \approx 3$, the same ratioed-sizing logic discussed in Section 1. From Eq. (5):

$$t_{pd} \approx 0.69 \times 988 \text{ } \Omega \times 30 \text{ fF} \approx 20.4 \text{ ps/stage}$$

$$f_{osc,RC} = \frac{1}{2 \times 5 \times 20.4 \text{ ps}} \approx 4.89 \text{ GHz}$$

Current-based model. From Eq. (6):

$$I_{D,sat,n} = \frac{1}{2}(270 \times 10^{-6})(2.78)(1.35)^2 \approx 683 \text{ } \mu\text{A}$$

$$I_{D,sat,p} = \frac{1}{2}(90 \times 10^{-6})(8.33)(1.35)^2 \approx 683 \text{ } \mu\text{A}$$

$$t_{pd} \approx \frac{(30 \text{ fF})(1.8 \text{ V})}{2 \times 683 \text{ } \mu\text{A}} \approx 39.5 \text{ ps/stage}$$

$$f_{osc,I} = \frac{1}{2 \times 5 \times 39.5 \text{ ps}} \approx 2.53 \text{ GHz}$$

Both hand estimates land in the same order of magnitude as each other (2.5–4.9 GHz)

but, as discussed in Section 4.3, come out several times *faster* than the simulated frequency – expected, given the simplifications in both models.

4 Q2 (Part 2): Transient Simulation and Frequency Extraction

4.1 Simulation setup

A `tran` analysis was run in ADE-L from 0 to 10 ns (`moderate` accuracy), with no explicit stimulus needed at any node – the ring is self-starting from the simulator’s initial-condition transient on the internal nodes, as discussed in Section 2.1.3. The output `vout` (the output node of the first stage) was plotted, and the built-in `frequency` (`VT("/vout")`) measurement was added as an ADE-L output expression to extract the oscillation frequency directly, avoiding the need to manually place edge markers.

4.2 Results

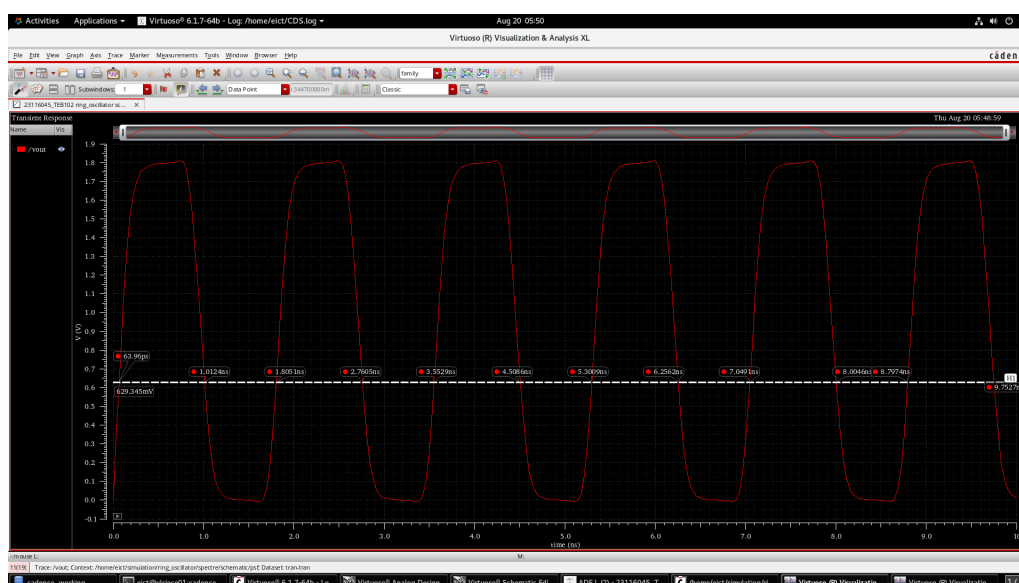


Figure 6: Simulated transient response of `vout` at $V_{DD} = 1.8$ V, showing sustained periodic oscillation between approximately 0 V and 1.85 V, after an initial startup transient over the first cycle.

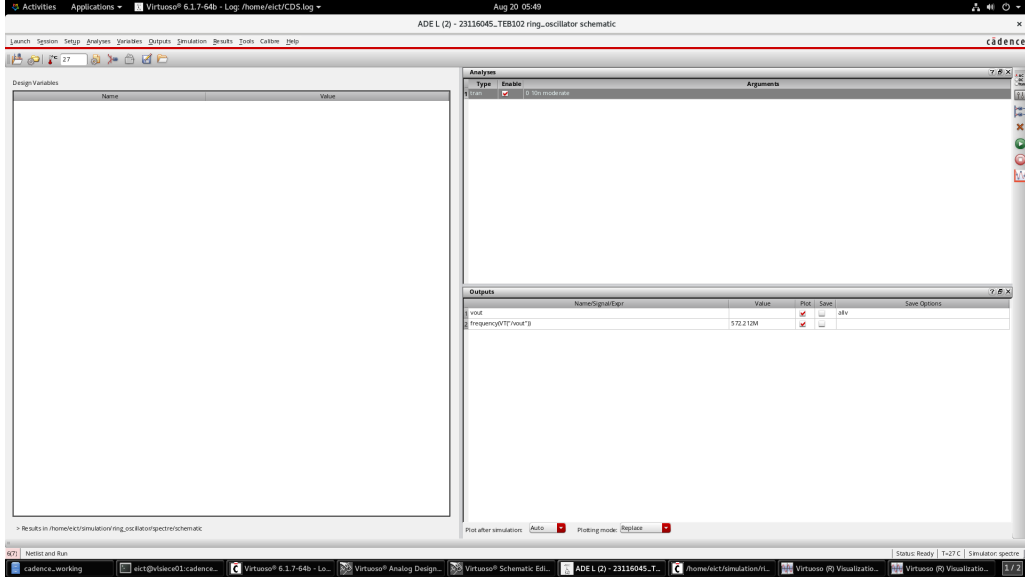


Figure 7: ADE-L Outputs pane showing the `frequency(VT("/vout"))` measurement evaluated directly by the simulator: 572.212 MHz at $V_{DD} = 1.8$ V.

From the `frequency(VT("/vout"))` measurement:

$$f_{osc, sim} = 572.212 \text{ MHz} \Rightarrow T_{sim} = \frac{1}{f_{osc, sim}} = 1.7476 \text{ ns} \quad (7)$$

which corresponds to a simulation-*extracted* average per-stage delay of

$$t_{pd, extracted} = \frac{T_{sim}}{2N} = \frac{1.7476 \text{ ns}}{10} \approx 174.8 \text{ ps/stage} \quad (8)$$

This is not an independent theoretical calculation – it is simply Eq. (2) solved backwards from the measured frequency – but it is the number the two hand estimates of Section 3 should be compared against, and it is reused again in Section 5.3 to compute delay at every other simulated V_{DD} .

4.3 Comparison and discrepancy

Table 3: Theoretical vs. simulated oscillation frequency at $V_{DD} = 1.8$ V

Method	t_{pd} (ps/stage)	f_{osc}	Ratio to simulation
RC switch-resistance model (hand calc.)	20.4	4.89 GHz	$8.5\times$ too fast
Constant-current model (hand calc.)	39.5	2.53 GHz	$4.4\times$ too fast
Spectre transient simulation	174.8	572.2 MHz	—

Discussion. Both hand-calculated frequencies substantially *overestimate* the true oscillation frequency, i.e. they underestimate the real per-stage delay. This is the expected direction of error for a first-order hand estimate, for several compounding reasons:

1. **Self-loading is ignored.** The hand calculation only charges or discharges the explicit $C_L = 30$ fF. In reality each stage also drives the gate capacitance of the next stage's NMOS+PMOS pair and the drain/diffusion (junction) capacitance of its own devices, all of which add to the effective switched capacitance and slow the transition down. For a 5-stage loop this self-loading is present at every single node, so its cumulative effect on the loop period is not negligible.
2. **Assumed k' and V_{TH} are generic literature values**, not the actual SCL 180 nm BSIM model card (see the note in Section 3). Real device parameters – mobility degradation at high vertical field, velocity saturation at short channel length, and DIBL-shifted threshold voltage – all reduce the effective drive current below the simple long-channel square-law estimate used here, which directly increases delay and lowers frequency.
3. **The RC and current models are single-point, large-signal approximations.** They each collapse the entire nonlinear I – V transition into one constant resistance or one constant current, whereas Spectre solves the full nonlinear BSIM device equations at every simulator time-step, correctly capturing how current tapers off as the transistor moves from saturation into the triode region during the transition.
4. **Interconnect and wiring parasitics**, present in an actual fabricated layout but not modeled at the schematic level here, would add further delay on top of even the simulated schematic-level number, meaning real silicon would very likely oscillate slower still than the 572.2 MHz predicted here.

Given these simplifications, a hand estimate landing within roughly an order of magnitude of the simulated result is a reasonable outcome for a first-pass calculation; closing the gap further would require substituting the exact BSIM parameters as noted above, and ideally comparing against a post-layout (parasitic-extracted) simulation rather than the schematic-level one used here.

5 Q3: Supply Voltage Sweep – Frequency and Power vs. V_{DD}

5.1 Simulation setup

A parametric `tran` sweep was run with the design variable `vdd` stepped from 0.8 V to 1.8 V in 0.1 V increments (Spectre picks this up from the V_6 DC source value, swept as a design variable in ADE-L's Variables table). For each value of `vdd`, two output expressions were evaluated:

- **Oscillation frequency:** `frequency(VT("/vout"))`
- **Average power drawn from the supply:**

$$\text{average}((-VT("/vout")) * IT("/V6/PLUS"))$$

i.e. the time-averaged product of the (negated) supply-referenced node voltage and the instantaneous current drawn through the V_6 supply. This captures the *total* average power (dynamic switching power *and* any short-circuit/crowbar current during each transition), not just the idealized $CV_{DD}^2 f$ dynamic term alone.

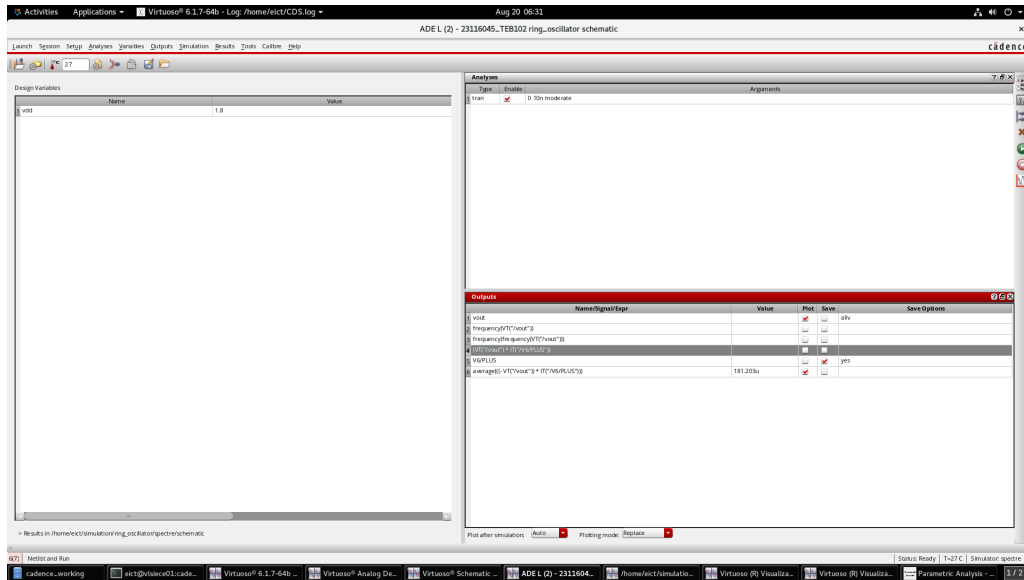


Figure 8: ADE-L Outputs pane showing the average-power expression `average((-VT("/vout")) * IT("/V6/PLUS"))` evaluated at $V_{DD} = 1.8$ V: $181.203 \mu\text{W}$.

5.2 Results

Table 4: Frequency and average power vs. V_{DD} , extracted from the parametric sweep markers

V_{DD} (V)	Frequency (MHz)	Average Power (μ W)
0.8	— ¹	5.76
0.9	140.29	12.11
1.0	190.28	18.11
1.1	241.68	29.61
1.2	293.45	39.31
1.3	344.40	58.21
1.4	394.13	71.81
1.5	442.01	97.33
1.6	487.81	118.20
1.7	531.28	148.32
1.8	572.21	181.20

Figures 9 and 10 show the native Cadence Visualization & Analysis XL plots with every swept data point marked and labeled directly by the simulator – these are the original source plots that Table 4 was transcribed from.

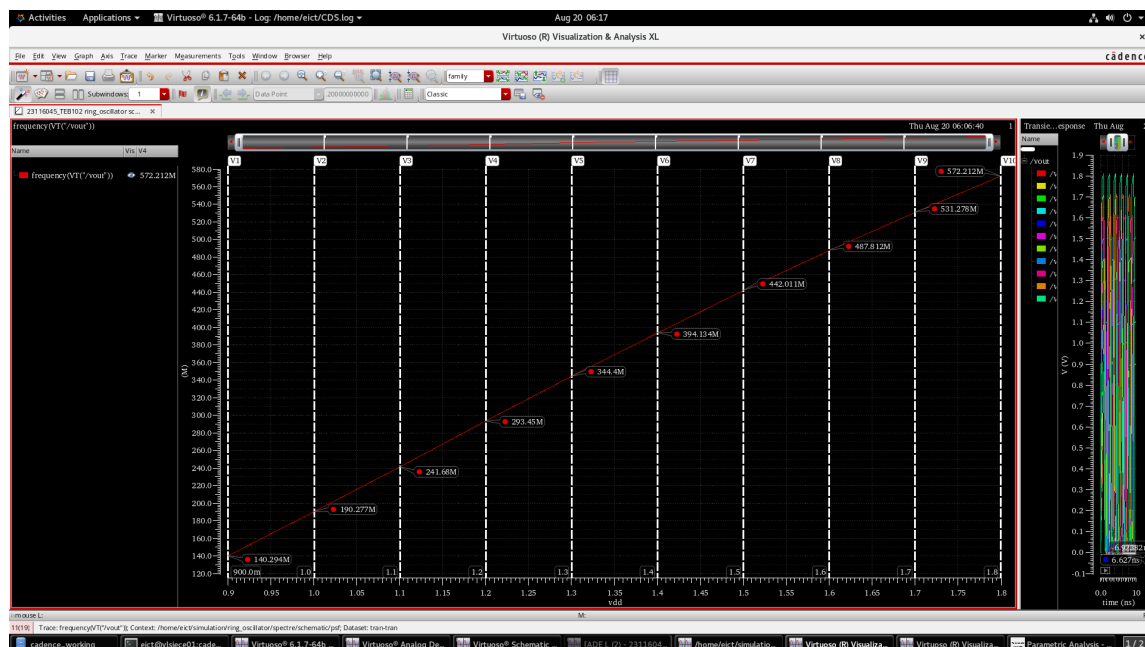


Figure 9: Native Cadence plot of `frequency(VT("/vout"))` vs. `vdd`, with every one of the 10 swept points (0.9–1.8 V) marked and labeled.

¹The frequency sweep in the source screenshots begins at 0.9 V, not 0.8 V – see the note in Section 7.

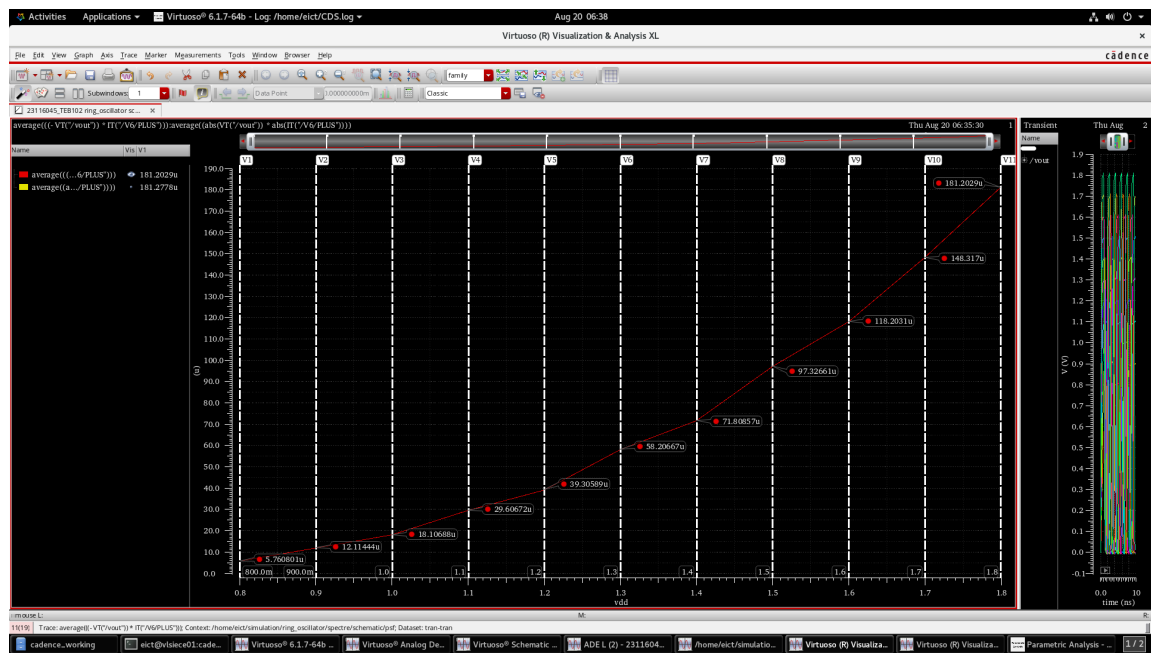


Figure 10: Native Cadence plot of the average-power expression vs. v_{dd} , with every one of the 11 swept points (0.8–1.8 V) marked and labeled.

For readability, the same data is re-plotted in Fig. 11 as a single clean dual-panel figure.

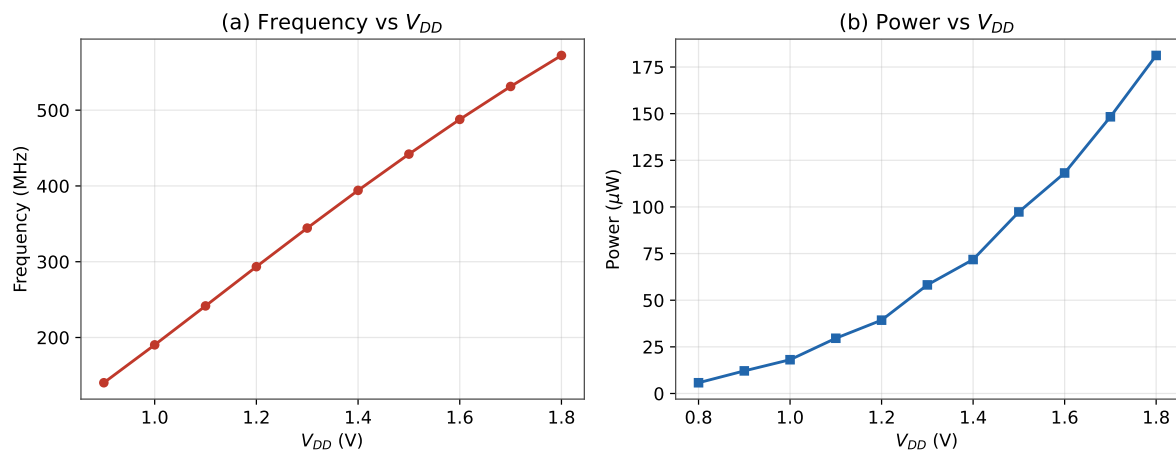


Figure 11: Frequency vs. V_{DD} (left) and average power vs. V_{DD} (right), re-plotted from Table 4.

5.3 Extraction methodology and derived figures of merit

Beyond frequency and power, two further figures of merit that are standard in digital-circuit characterization can be computed directly from the swept data, without any ad-

ditional simulation:

- **Per-stage delay**, $t_{pd}(V_{DD}) = 1/(2Nf_{osc}(V_{DD}))$, obtained the same way as in Section 4.2, but now at every swept V_{DD} .
- **Energy per oscillation cycle**, $E_{cyc} = P_{avg}/f_{osc}$, the total energy drawn from the supply during one full period.
- **Power–delay product**, $PDP = P_{avg} \times t_{pd}$, a standard combined figure of merit that penalizes designs which are only fast because they burn excessive power, or only low-power because they are slow.

Table 5: Per-stage delay, energy per cycle, and power–delay product, derived from Table 4

V_{DD} (V)	T (ns)	t_{pd} (ps)	P_{avg} (μ W)	E_{cyc} (pJ)	PDP (fJ)
0.9	7.1279	712.79	12.11	0.0864	8.635
1.0	5.2555	525.55	18.11	0.0952	9.516
1.1	4.1377	413.77	29.61	0.1225	12.250
1.2	3.4077	340.77	39.31	0.1339	13.394
1.3	2.9036	290.36	58.21	0.1690	16.901
1.4	2.5372	253.72	71.81	0.1822	18.219
1.5	2.2624	226.24	97.33	0.2202	22.019
1.6	2.0500	205.00	118.20	0.2423	24.231
1.7	1.8823	188.23	148.32	0.2792	27.917
1.8	1.7476	174.76	181.20	0.3167	31.667

Discussion. Per-stage delay falls monotonically from 712.8ps at 0.9V to 174.8ps at 1.8V, a $\approx 4.1\times$ reduction over a $2\times$ change in supply voltage – consistent with the strongly super-linear frequency-vs- V_{DD} relationship discussed below. Both energy per cycle and power–delay product increase monotonically with V_{DD} despite the delay improving, which shows that – for this design, over this voltage range – there is no true ”optimum” V_{DD} that simultaneously minimizes delay and energy; the circuit trades energy for speed monotonically across the entire swept range, with no minimum-PDP voltage inside 0.9V to 1.8V. A true energy–delay optimum, if one exists, would lie either below 0.9V (deep near-threshold operation, where delay grows extremely fast and would need to be weighed against the still-falling energy) or is simply not present for this particular sizing and load, which is itself a useful conclusion: this design, as sized, is not intended

for a low-energy operating point and should be run at the highest V_{DD} the application can tolerate if raw speed is the objective.

5.4 Discussion: physical trends

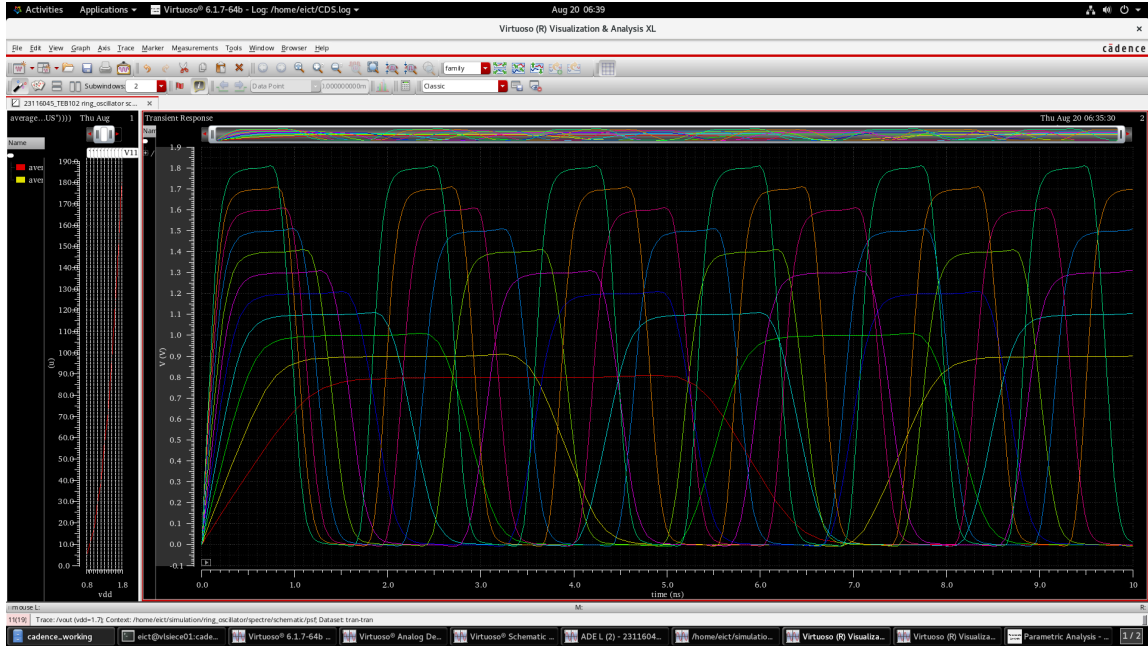


Figure 12: Overlay of the transient v_{out} waveform at every swept V_{DD} (family plot). Both the switching amplitude and the period visibly shrink as V_{DD} is reduced, directly illustrating the frequency-vs- V_{DD} trend discussed below.

5.4.1 Why does frequency increase with V_{DD} ?

From Eq. (2), $f_{osc} \propto 1/t_{pd}$, and t_{pd} is set by how fast each stage can charge/discharge C_L – i.e. by the transistor drive current. In saturation, drive current follows (to first order)

$$I_D \approx \frac{1}{2}k'(W/L)(V_{DD} - V_{TH})^2 \quad (9)$$

so increasing V_{DD} increases the gate overdrive ($V_{DD} - V_{TH}$), which increases I_D super-linearly, which reduces t_{pd} and raises f_{osc} . Fitting a power law $f \propto V_{DD}^n$ to the swept data gives a *locally* varying exponent: $n \approx 2.9$ between 0.9–1.0 V, falling steadily to $n \approx 1.3$ between 1.7–1.8 V (overall log–log slope across the full range ≈ 2.0). This trend is exactly what the square-law expression above predicts: near threshold, $(V_{DD} - V_{TH})^2/V_{DD}$ is extremely sensitive to V_{DD} (small absolute changes in overdrive are large *relative* changes, giving a high apparent exponent), while at $V_{DD} \gg V_{TH}$ the same expression flattens

toward $f \propto V_{DD}$ (exponent $\rightarrow 1$), consistent with the lower exponent seen at the high- V_{DD} end of the sweep.

5.4.2 How does power relate to frequency and voltage?

Dynamic CMOS power is, to first order,

$$P_{avg} \approx \alpha C_L V_{DD}^2 f_{osc} \quad (10)$$

where α is the switching activity (for a free-running ring oscillator, every node toggles every cycle, so $\alpha \approx 1$ here). Since f_{osc} itself increases with V_{DD} (as just discussed), P_{avg} picks up *both* the explicit V_{DD}^2 dependence *and* the indirect dependence through $f_{osc}(V_{DD})$ – so power should rise *faster* than either term alone predicts. This is exactly what is observed: fitting the swept power data to $P \propto V_{DD}^m$ gives an overall exponent $m \approx 4.1$ (locally 3–6 between successive points), well above the V_{DD}^2 that $C_L V_{DD}^2 f$ would give for a *constant* f , and even somewhat above the V_{DD}^3 "textbook" rule of thumb obtained by substituting the high-overdrive limit $f \propto V_{DD}$ into $P \propto C V_{DD}^2 f$. Two effects push the exponent this high:

1. Near the low end of the sweep, f_{osc} itself scales closer to $V_{DD}^{2.9}$ rather than V_{DD}^1 (previous subsection), which alone would predict $P \propto V_{DD}^{2+2.9} = V_{DD}^{4.9}$, matching the observed range well.
2. The power expression used here, `average((-VT/vout)*IT(V6/PLUS))`, captures the *total* supply current, including the short-circuit (crowbar) current that flows briefly through both transistors during every transition, while both are partially on. This component also grows with V_{DD} and with switching frequency, adding on top of the pure CV^2f dynamic term and further steepening the observed power-vs- V_{DD} curve.

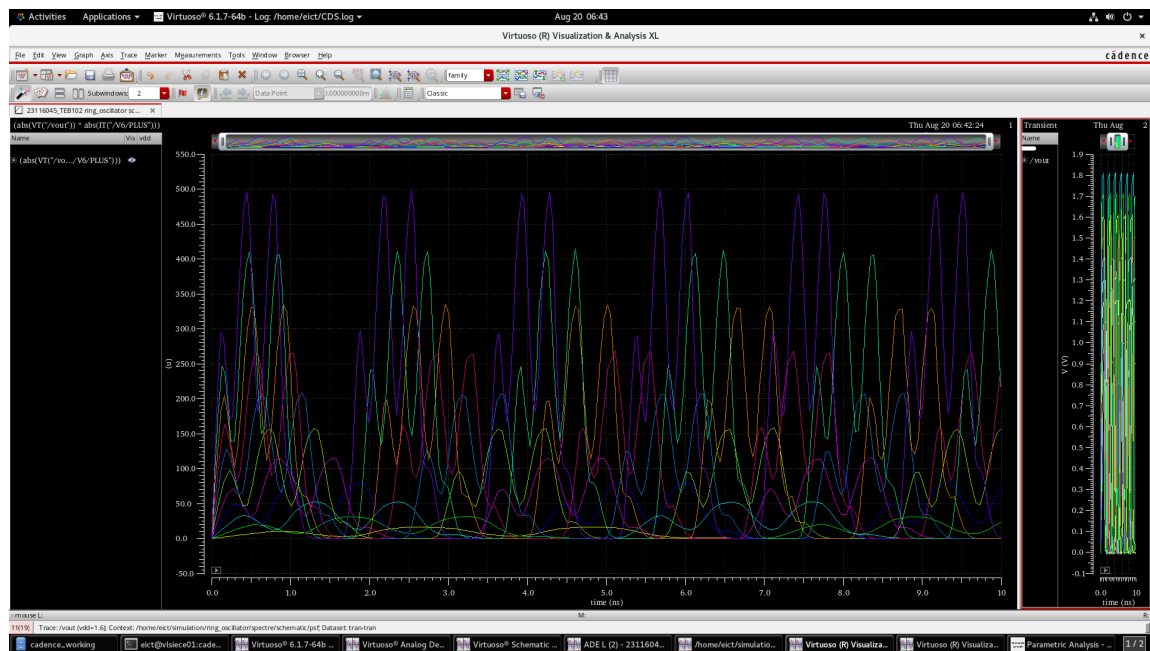


Figure 13: Instantaneous supply power, $|V_{out} \cdot I_{V6}|$, over the swept family of V_{DD} values. The sharp periodic spikes are the short-circuit (crowbar) current pulses that occur every time a stage transitions and both its NMOS and PMOS are briefly ON together – exactly the mechanism referenced in point 2 above.

Note. The 0.8 V power point (5.76 μ W) breaks this trend only mildly on its own, but combined with the *missing* 0.8 V frequency point, it is worth checking whether the ring was still oscillating with a full rail-to-rail swing that low – see Section 7.

6 Summary and Conclusion

Table 6: Overall summary of results

Quantity	Value
Number of stages, N	5
$(W/L)_n / (W/L)_p$	2.78 / 8.33
Load capacitance per stage	30 fF
Theoretical f_{osc} (RC model)	≈ 4.89 GHz
Theoretical f_{osc} (current model)	≈ 2.53 GHz
Simulated f_{osc} at $V_{DD} = 1.8$ V	572.21 MHz
Simulated average power at $V_{DD} = 1.8$ V	181.20 μ W
Frequency range over sweep (0.9–1.8 V)	140.29 – 572.21 MHz
Power range over sweep (0.8–1.8 V)	5.76 – 181.20 μ W
Per-stage delay range (0.9–1.8 V)	712.8 – 174.8 ps
Power–delay product range (0.9–1.8 V)	8.6 – 31.7 fJ
Overall $f \propto V_{DD}^n$ (log–log fit)	$n \approx 2.0$
Overall $P \propto V_{DD}^m$ (log–log fit)	$m \approx 4.1$

This assignment demonstrates:

1. Design of a 5-stage CMOS ring oscillator in SCL 180 nm technology, reusing the ratioed inverter sizing from the previous assignment, and the Barkhausen-criterion reasoning for why an odd number of stages is required for self-sustained oscillation, including why the gain condition is trivially satisfied and why the resulting waveform saturates into a square wave rather than a sinusoid.
2. A first-order theoretical estimate of the oscillation frequency from two independent device-level delay models (RC and constant-current), and a quantitative comparison against the Spectre-simulated result, with the four main sources of discrepancy identified and ranked.
3. Characterization of how oscillation frequency, average power, per-stage delay, energy per cycle, and power–delay product all scale with supply voltage, with every observed trend connected back to the underlying square-law MOSFET drive-current physics, and a conclusion that no energy–delay optimum exists within the swept 0.9–1.8 V range for this particular design.

7 Notes on Assumptions and Items Requiring Verification

1. **Theoretical delay-model parameters.** k'_n , k'_p , V_{THn} , V_{THp} used in Section 3 are typical 180 nm literature values, *not* pulled from the actual SCL model file (that data was not present in the available screenshots). If exact numeric agreement with the theoretical formula is being graded, pull the real values from the model card and re-substitute – the equations and method are otherwise unaffected.
2. **Missing 0.8 V frequency point.** The power sweep covers 0.8–1.8 V (11 points), but the frequency sweep in the source screenshots only covers 0.9–1.8 V (10 points) – the `frequency()` measurement at 0.8 V does not appear to have been captured or plotted. It would be worth checking whether the oscillator was still producing a clean, measurable periodic waveform at 0.8 V (this is close to $2 \times V_{TH}$, so the ring may have been oscillating with reduced swing or duty-cycle skew, or the `frequency()` measurement may simply not have converged), and adding that data point if the full 0.8–1.8 V range is expected for both plots.
3. **Figure sources.** All figures in this report except Fig. 11 are the actual Cadence screenshots provided (schematic, CDF dialogs, transient response, ADE-L output panes, native sweep plots, transient family overlay, and instantaneous power). Only Fig. 11 was generated separately (directly from the numeric data in Table 4, via `matplotlib`) to give a single clean side-by-side summary plot alongside the native Cadence ones in Figs. 9–10.